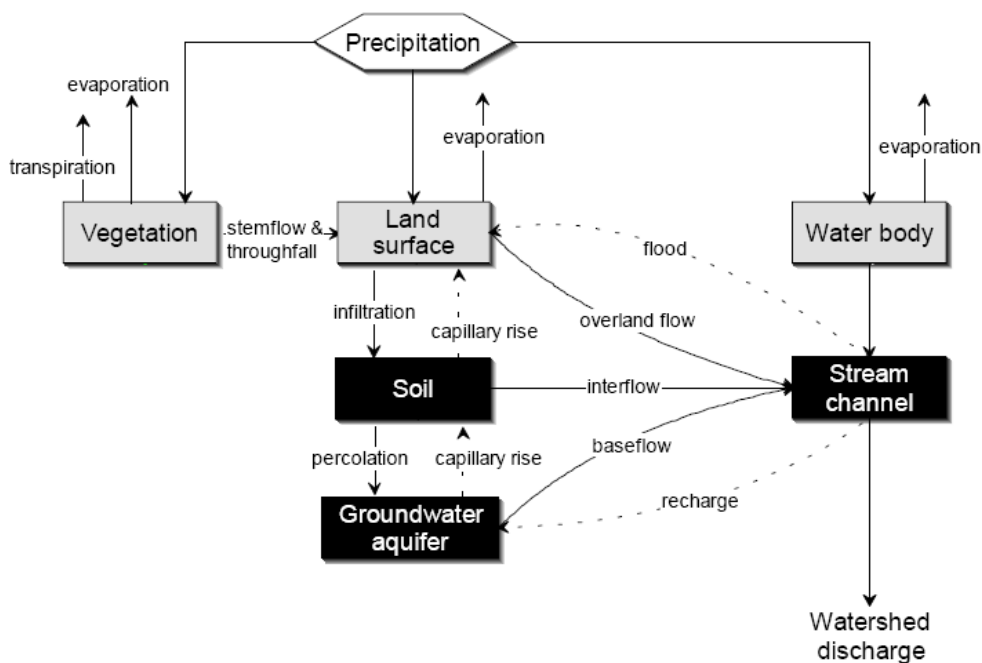


Topic	Surface Water Modelling	
Module	1.	Hydrological Modelling
Exercises		Solutions to Exercises

Exercise 1

Construct a hydrological model concept by selecting an adequate process representation for the processes shown in the Figure shown below. Make sure to define the purpose of the hydrological model first. How would your model concept change if the purpose changes? List all model parameters and discuss potential sources for these model parameters. Which data are needed to run the model that you have constructed?



Solution

Here one example is given.

Model purpose

Design flood estimation for design of flood protection measures in an urban environment.

How would the model concept change if the purpose changes?

The number of processes that must be considered depends on the purpose of the model. For example, if the purpose is to be able to determine how much water can be taken from a river for irrigation and municipal water supply, then a continuous hydrological model must be developed to determine daily flows.

Model concept, parameters, sources data requirements

<i>Process</i>	<i>Process representation</i>	<i>Model parameters and sources</i>	<i>Data requirements</i>
Rainfall	Synthetic hyetograph	n/a	Observed data and DDF
Infiltration	SCS-CN	CN, initial loss; Source: Literature and local information	Soil, land use and catchment characteristics area (pervious and impervious)
Runoff concentration	SCS Unit Hydrograph	Time of concentration, area; Sources: Calculated from data, local information	Catchment characteristics: length, slope

Output: Peak flow, volume and duration of design flood.

Exercise 2

Develop a hydrological model using HEC-HMS.

Solution

See computer lab.

Exercise 3

Use airGRteaching (<https://sunshine.inrae.fr/app/airGRteaching>) to explore manual and automatic calibration of a hydrological model. Also explore using different types of plots to assess the model calibration exercise.

Solution

See class recording.

Exercise 4

Use the HMS model to assess the impact of the initial storage on the hydrograph dynamics.

Solution

See computer lab.

Exercise 5

Calibrate the HEC-HMS model using observed data.

Solution

See computer lab.

Exercise 6

Validate the HEC-HMS model using observed data.

Solution

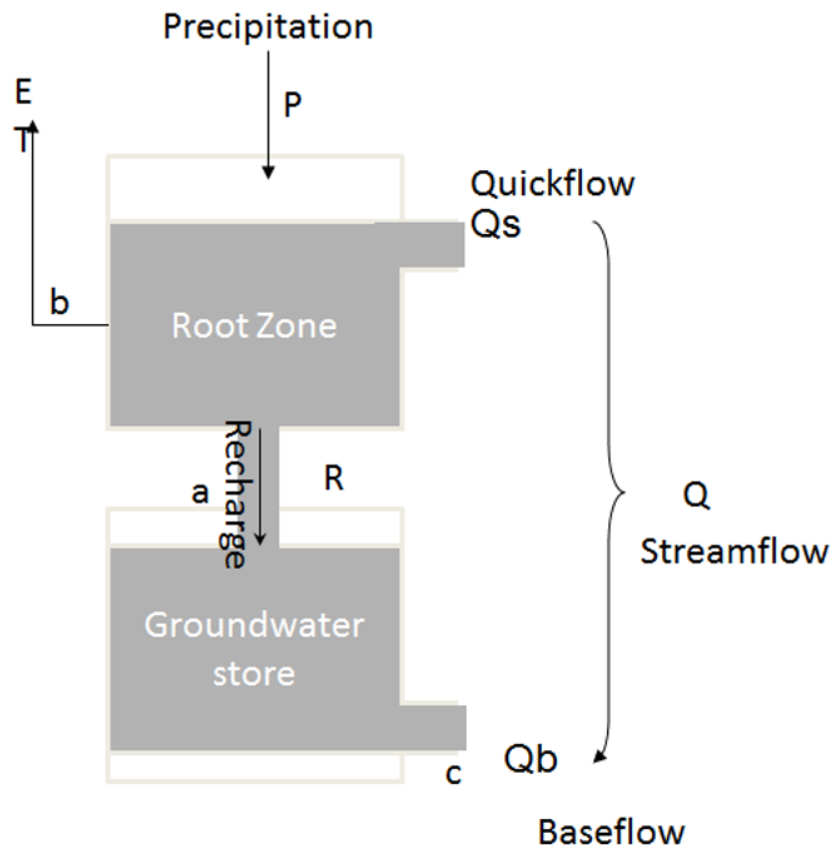
See discussion in class and in the computer lab.

Additional Exercise introduced in class

Develop a 3-parameter model, the 'abc model'.

Solution

To partition precipitation input into evapotranspiration loss, groundwater storage and streamflow, three parameters a, b and c are employed. In a first step a schematic representation of such a hydrological modelling system is required.



If precipitation at time t is represented by P_t , then combined infiltration and deep percolation is

$$R_t = a P_t$$

and evapotranspiration is

$$ET_t = b P_t,$$

where a and b represent the fraction of rainfall which builds the sum of infiltration and deep percolation through the soil, and the sum of evaporation (from open water bodies, the soil surface and the surface of plant) and transpiration (through the plants), respectively.

The remaining component of rainfall,

$$P_t - R_t - ET_t = (1 - a - b) P_t,$$

results in surface runoff to the stream channel. Groundwater storage at time t is S_t , and the groundwater outflow to the stream channel is a fixed fraction cS_{t-1} of groundwater storage in the previous period. Finally, streamflow, Q_t , is given as the combination of surface and groundwater inputs in addition to model error ε_t

$$Q_t = (1 - a - b) P_t + c S_{t-1} + \varepsilon_t ,$$

so the quickflow component is

$$Q_{s,t} = (1 - a - b) P_t$$

and the baseflow component is

$$Q_{b,t} = c S_{t-1}.$$

The groundwater storage at the end of the time step t is derived by continuity as previous groundwater storage S_{t-1} less groundwater outflow plus deep percolation and model error v_t

$$S_t = a P_t + (1-c) S_{t-1} + v_t$$

It can be seen that the three-parameter model does, in the present form, not include a soil moisture component.

The index t describes the time step ($t=1,2,\dots,N$). Streamflow, precipitation and storage are measured in the same units (e.g. length over time or volume over time), so that the additive relations derived are dimensionally homogeneous.

For this model, the following constraints emerge:

Since the parameters represent fractions, these have upper limits: $0 \leq a, b, c \leq 1$

Since infiltration & deep percolation and evapotranspiration combined cannot exceed total precipitation: $0 \leq a + b \leq 1$

Physics demands that precipitation and storage cannot be negative: $P_t, S_t \geq 0$